Mohammad Elsharqawy

LLM Systems & AI Infrastructure

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Education

Tanta University Tanta, Egypt
B.Sc. in Computer Engineering and Automatic Control *Oct. 2018 - Jun. 2023*
- Ranked 4th out of 150+ students; graduated Excellent with Honors (the highest possible grade).

Research Software Contributions

libCacheSim - High-Performance Cache Library and Simulator 2026
MultiQueue (MQ) eviction algorithm in C | Project led by Prof. Juncheng Yang (Harvard University) *PR #318 | issue #140*
- Implemented the MultiQueue eviction algorithm (Zhou, Philbin & Li, USENIX ATC 2001), approximately 550 LOC; validated correctness through an LRU-equivalence reduction at n-queue=1, regression tests on ground-truth miss counts, and evaluation against seven baselines across traces and cache sizes.

SREGym - Benchmark for Agentic Site Reliability Engineering (UIUC) 2026
Reproducing production outages as evaluable Kubernetes faults *PR #774 | PR #791*
- GKE clusters run out of pod IP addresses, preventing new workloads from starting: reproduced the failure in Kubernetes using Calico IPAM with strict affinity and added cleanup logic for deterministic, repeatable runs.
- Reproduced a Fastly-style configuration-triggered latent bug in which a ConfigMap activates a dormant application error path, causing HTTP 500 responses while all Kubernetes pods remain Running; added deterministic fault activation and mitigation checks.

Research Experience

Research Intern - System Reliability & Agentic SRE May 2026 - Aug. 2026
University of Illinois Urbana-Champaign (UIUC), advised by Prof. Tianyin Xu *Remote*
- Investigated how LLM-based SRE agents diagnose and remediate realistic failures in cloud-native systems.
- Studied how production incidents can be translated into reproducible benchmark scenarios while preserving causal structure and diagnostic difficulty; designed failure scenarios and evaluated agents on them.
- Identified limitations in cross-namespace reasoning and configuration-to-symptom correlation, and analyzed which mechanisms from large-scale postmortems faithfully project to benchmark environments.

Research Assistant - Robustness under Distribution Shift Oct. 2025 - Present
Tanta University, advised by Prof. Mohamed T. Faheem Saidahmd *Tanta, Egypt*
- Investigating domain generalization for diabetic retinopathy grading across camera hardware, acquisition protocols, and patient populations.

Selected Projects

LLM Inference Engineering - Systems from First Principles repository
- Studied LLM inference internals including attention, KV-cache behavior, model-weight movement, and the distinction between compute-bound prefill and memory-bound autoregressive decoding.
- Built PyTorch experiments for attention FLOPs, memory traffic, arithmetic intensity, batching, and sequence-length scaling; used roofline-style reasoning to explain inference bottlenecks and GPU utilization.
- Benchmarked representative matrix-multiplication workloads: decode rose from about 0.2 TFLOPS at batch size 1 to 6.1 TFLOPS at batch size 32; long-sequence prefill reached about 23 TFLOPS.

Teaching Experience

Teaching Assistant Sep. 2024 - Present
Tanta University - Computer Engineering / Artificial Intelligence Departments *Tanta, Egypt*
- Lead instruction, labs, and assessment for Operating Systems, Data Structures & Algorithms, Artificial Intelligence, Machine Learning, OOP, and programming courses.
- Coached 11 teams at the Egyptian Collegiate Programming Contest 2023, including six as head coach.

Professional Experience

AI, Data Science and Cloud Engineering Trainee - Digital Egypt Builders Initiative Jan. 2024 - Sep. 2024
- Selected for a national digital-talent program with intensive training across AI, data science, and cloud engineering.

Software Engineering Intern - Ithra Services Ltd. Jul. 2022 - Sep. 2022
- Built REST APIs with MongoDB and JWT authentication; containerized services with Docker and automated AWS EC2 deployments using GitHub Actions and Ansible.

Honors and Awards

- 4th out of 150+ in graduating class, Tanta University (2023); Top Students Financial Award for five consecutive years (2019-2023).
- Finalist, ACM-ICPC Egyptian Collegiate Programming Contest (2021, 2022); 2nd place, Tanta University Collegiate Programming Contest (2021, 2022).

Technical Skills

Programming: Python, C/C++, Go, Java, JavaScript, Bash, VHDL
Systems & Infrastructure: Linux, Kubernetes, Docker, Terraform, Ansible, Jenkins, GitHub Actions, Argo CD, Helm, AWS, Calico CNI
ML & AI Systems: PyTorch, Transformers, LLMs, RAG, LoRA, fine-tuning, quantization, ONNX Runtime, MLflow
Observability: Prometheus, Grafana, Elasticsearch, Fluent Bit, Kibana, Jaeger
Data & Backend: MySQL, MongoDB, Pinecone, Flask, Node.js, RESTful APIs, RPC
Languages: Arabic (native); English (professional working proficiency)